

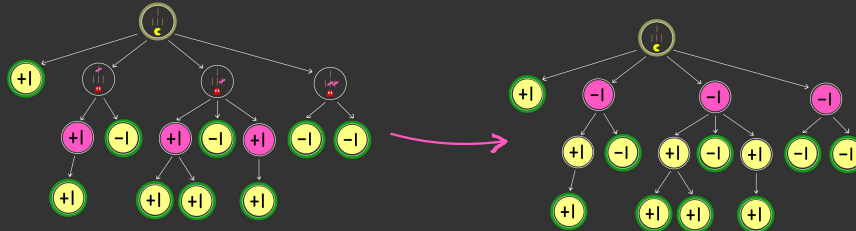
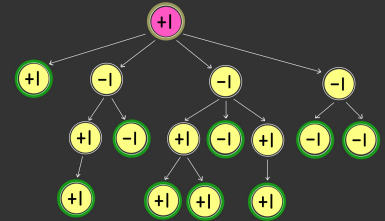
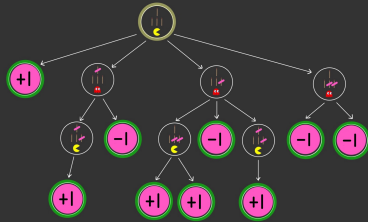
evaluation
functions

CSCI
270

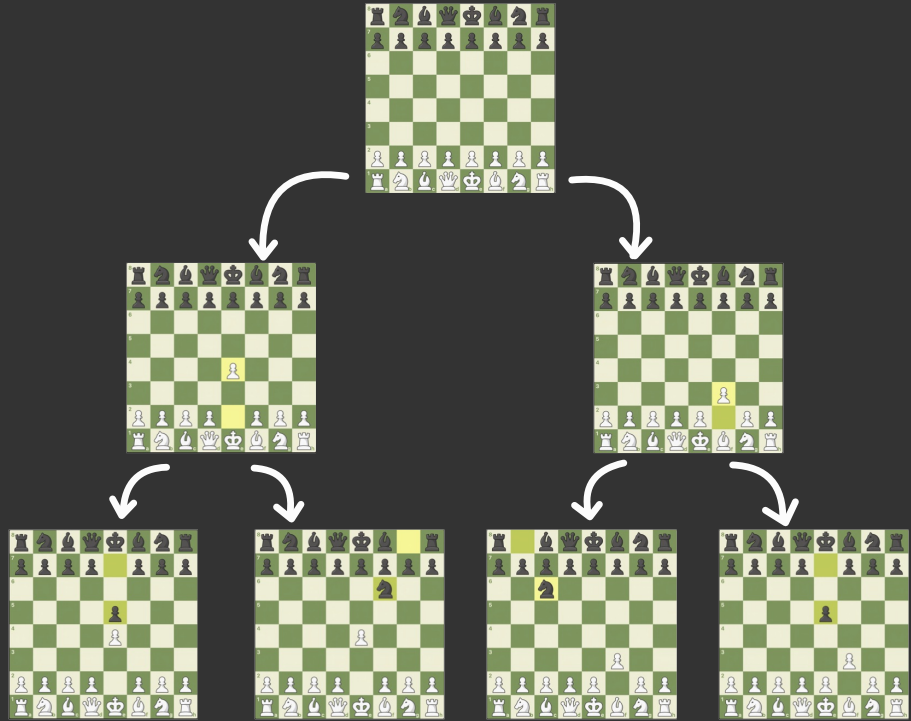
essentially minimax is a "bottom-up" algorithm,
computing values from the leaves up to the root

```

minimax( $q, p$ ):
  if  $q \in F$ :
    return  $U(q, p)$ 
  else if  $p(q) = p$ :
    return  $\max_{q'} \{ \text{minimax}(q, p) \mid (q, \sigma, q') \in \Delta \}$ 
  else:
    return  $\min_{q'} \{ \text{minimax}(q, p) \mid (q, \sigma, q') \in \Delta \}$ 
  
```



but what if the
search tree is
too **big** to get
to the leaves?



at some point we need to

cut off

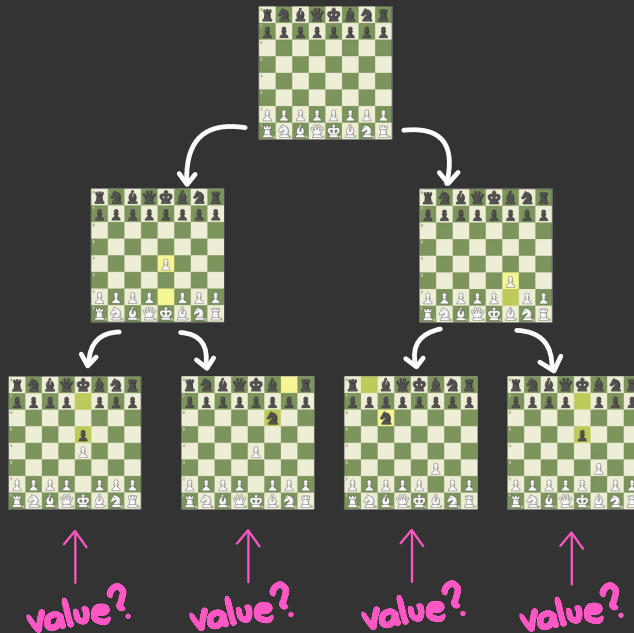
the minimax search and

estimate

the "goodness" of a state



The best upper bound found for the number of chess positions
is 7728772977965919677164873487685453137329736522



How many legal states of chess exists?

▲ You should clarify whether you wish to differentiate between positions based on en passant, castling, and whose side it is to move. Francis Labelle and others have used the term "chess position" to indicate a board state including the above information, and "chess diagram" to indicate a board state not including the above information, i.e. just what pieces are on the board and where. In neither case is information for drawing rules like the 50-move rule or the triple repetition rule included.

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✓
🕒

The best upper bound found for the number of chess position is 7728772977965919677164873487685453137329736522, or about 7.7×10^{45} , based on a complicated program by John Tromp; according to him, better documentation is required in order for the program to be considered verifiable. He also has a much simpler program that gives an upper bound of 4.5×10^{46} .

For chess diagrams, Tromps simpler program gives an upper bound of about 2.2×10^{46} ; he does not say what bound is obtained by the complicated program, but it is probably a little less than half (since side to move doubles the bound, whereas castling and en passant add relatively little), so likely about 3.8×10^{45} . More information at [Tromp's website](#)

The best bound published in a journal was obtained by Shirish Chinchalkar in "An Upper Bound for the Number of Reachable Positions". I do not have access to this paper, but according to Tromp it is about $10^{46.5}$. Although it refers to "positions", it could very easily be a bound for the number of diagrams.

As for lower bounds, they are much more difficult, since a given position could be illegal for very subtle reasons. Wikipedia has claimed that the number of positions is "between 10^{45} and 10^{47} ", but I think it is unlikely that the lower bound has been proven.

I would guess that the actual number of diagrams is between 10^{44} and 10^{45} , but this is purely speculation.

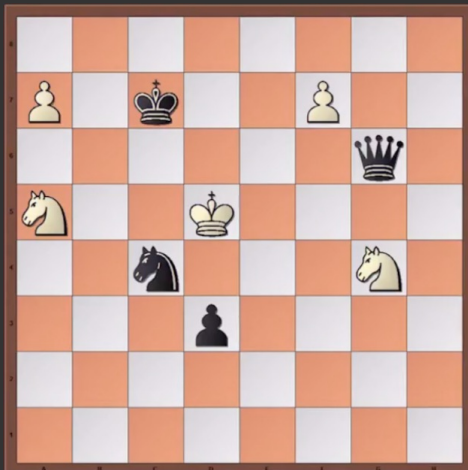
math.stackexchange.com

how might we
estimate the
goodness of a
non-final state?

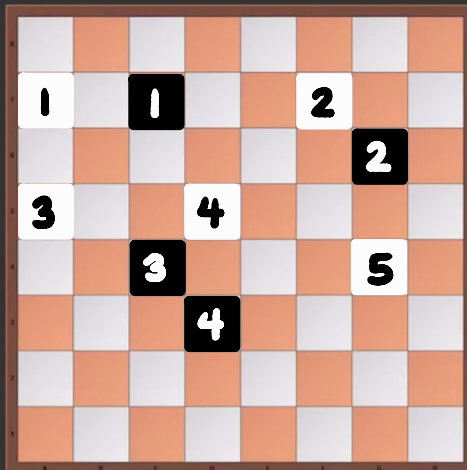


black to move

idea 1: count difference in
number of pieces



black to move



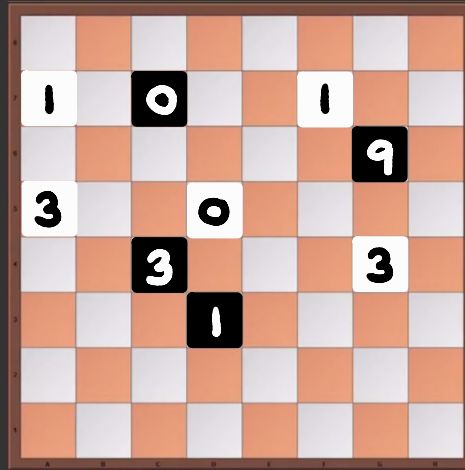
black to move

$$\text{evaluation} = 4 - 5 = -1$$

idea 2: count difference in
value of pieces



black to move

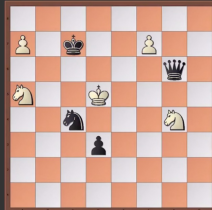


black to move

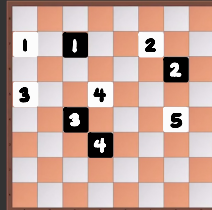
$$\text{evaluation} = 13 - 8 = 5$$

a function that estimates the goodness of a non-final state is called an **evaluation function**

idea 1: count difference in **number** of pieces



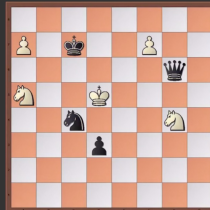
black to move



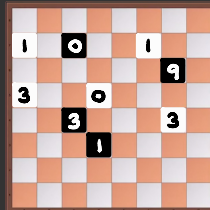
black to move

$$\text{evaluation} = 4 - 5 = -1$$

idea 2: count difference in **value** of pieces



black to move

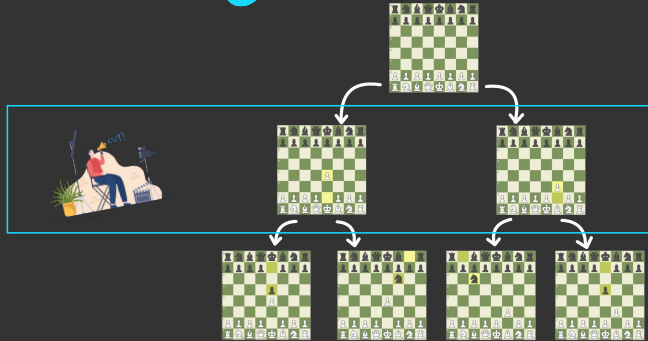


black to move

$$\text{evaluation} = 13 - 8 = 5$$

the **search depth** at which you cut off minimax and apply the evaluation function is called the number of "**plys**"

1-ply minimax



2-ply minimax

